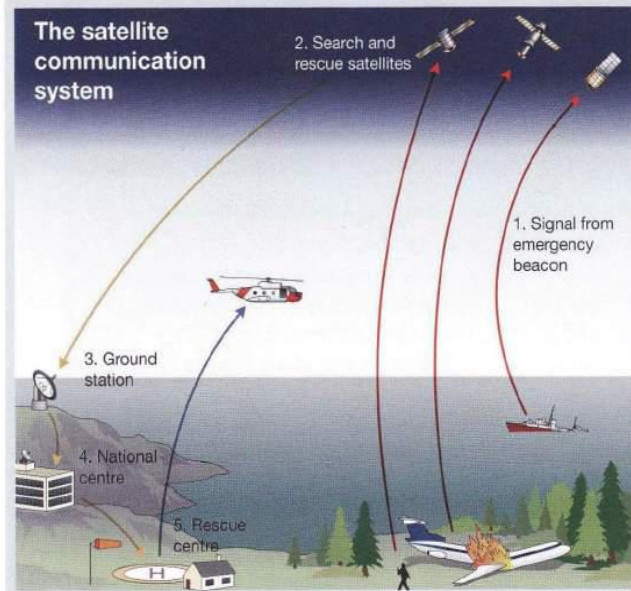


2 Transmission

Start here 1 Complete this description of how a satellite communication system works, using the correct form of the verbs in the box.

receive convert detach activate carry out transmit locate



If a plane crashes, or a ship sinks, the survivors try to (1) _____ their personal emergency beacons manually. In addition, an automatic beacon (2) _____ itself from the plane or ship and switches on automatically. The beacon then (3) _____ a signal to one or more satellites. The satellites (4) _____ the beacon's transmission and then send the beacon's signal to their ground station. The ground station then processes the satellite signals (that is, it (5) _____ the signals into useful data), and then passes on the data about the beacon to a national centre. The national centre forwards this data to the rescue centre nearest to the crashed plane or sinking ship. The rescue centre then (6) _____ the beacon and sends out a rescue team, which then (7) _____ the rescue.

Listening 2 04 Listen to this discussion and check your answers to 1.

Reading 3 Part of this text is missing. Write the letters of phrases A–G below in the correct spaces. Use the illustration in 1 to help you.

The Cospas-Sarsat system is an international search and rescue system which consists of a network of satellites in space, and control centres on Earth.

The components of the system are:

- radio beacons, which (1) _____
- satellites, which (2) _____
- ground stations, where (3) _____
- national centres, from where (4) _____
- rescue teams, who (5) _____

The system uses two types of satellite:

- satellites in geostationary Earth orbit (GEO), which (6) _____
- satellites in low-altitude Earth orbit (LEO), which (7) _____



geo- = Earth
Geostationary satellites move at the same speed and in the same direction as the Earth. When we observe them, they seem to be stationary or not moving.

- A are closer to the earth and cover polar regions.
- B information about the emergency is sent to the rescue teams.
- C are at a high altitude and cover a wide area.
- D transmit 406 MHz signals in an emergency.
- E signals from the satellites are processed.
- F pick up the signals from the beacons.
- G receive the information and carry out the search and rescue.

